# Strategic Problem Solving with Prolog

# Learning Outcomes

by the **End** of the **Lecture**, **Students** that **Complete** the **Recommended Exercises** should be **Able** to:

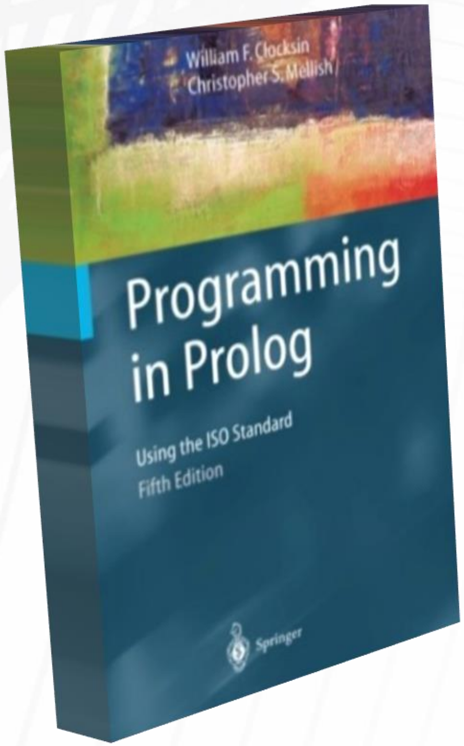
**Model** Finite State Machines with Prolog

**Use** Prolog for Graph Representation

**Explain** the "Fallen Frog" Problem

**Solve** "Fallen Frog" using Prolog

# Optional Reading Assignment




**Read** the following **Chapter(s)**:

None

# "Fallen Frog"

*A frog is at the bottom of a 30 meter well.*
*Each day he summons enough energy for one three meter leap up the well.*
*(i.e., After the first leap his hind legs are exactly three meters up the well.)*
*Exhausted, he then hangs there for the rest of the day.*
*At night, while he is asleep, he slips two meters backwards.*
*His hind legs must clear the well for him to escape.*

**How Long** does it take the **Frog** to **Escape** the **Well**?

# "Fallen Frog"

5

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*His hind legs must clear the well for him to escape.*

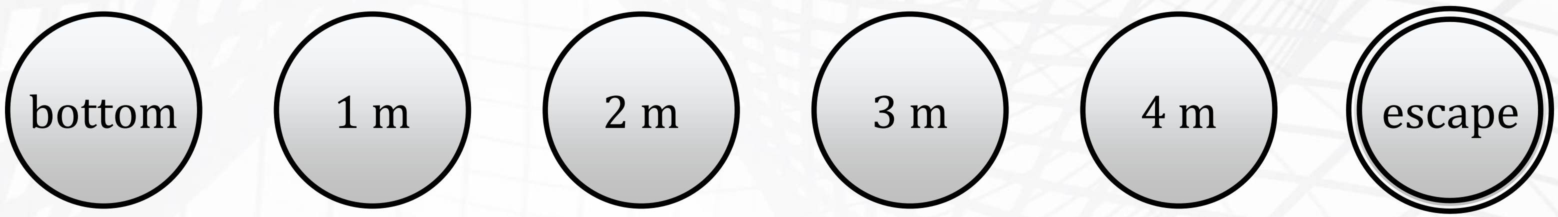
**How Long** does it take the **Frog** to **Escape** the **Well**?

while **Solving** such a **Problem**, the **Progress** (at any given moment) can be **Summarized** by **Describing** the **Current State** of the **Problem**

for many problems, there is a **Finite Number** of **Possible States**, so a **Finite State Machine** can be **Useful** for **Solving** such **Problems**

# "Fallen Frog"

**What** are the **States** for the 5m variant?



# "Fallen Frog"

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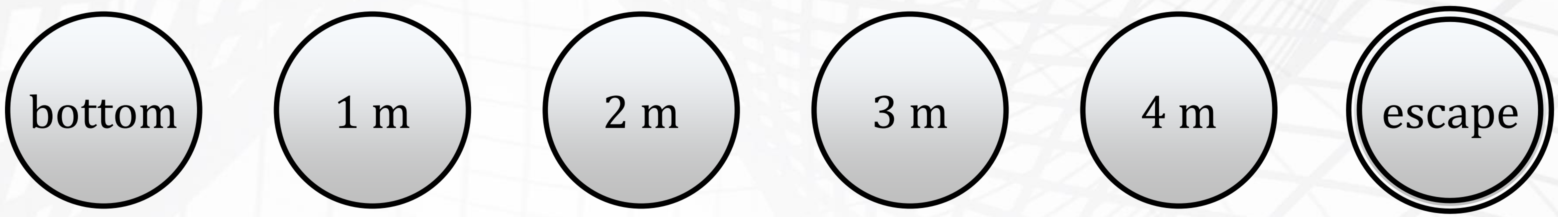


**What** about the **Time of Day**?

# "Fallen Frog"

~~**What** are the **States** for the 5m variant?~~

~~bottom | 1 m | 2 m | 3 m | 4 m | escape~~

**What** about the **Time of Day**?

| | | | | | |
|---|---|---|---|---|---|
| bottom day | 1 m day | 2 m day | 3 m day | 4 m day | escape day |
| bottom night | 1 m night | 2 m night | 3 m night | 4 m night | escape night |

**What** are the (some* of the) **Transitions**?

** n.b., it is possible for leap or rest from each of the different states*

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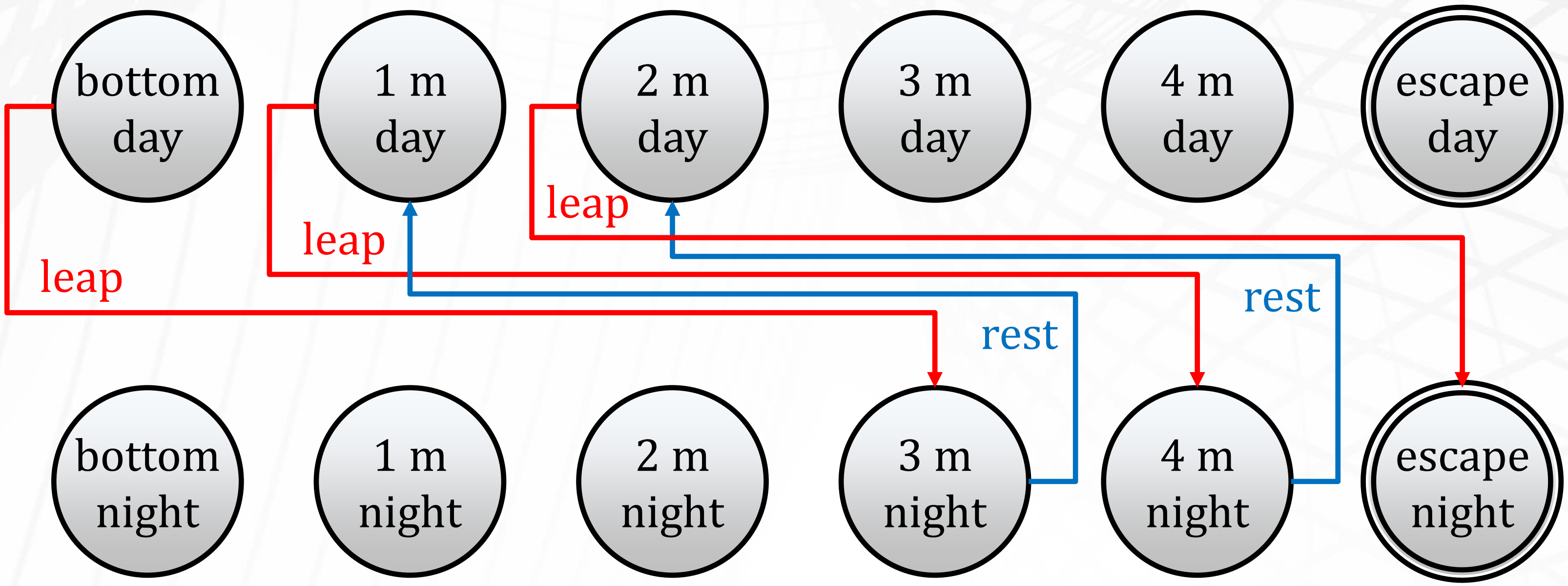
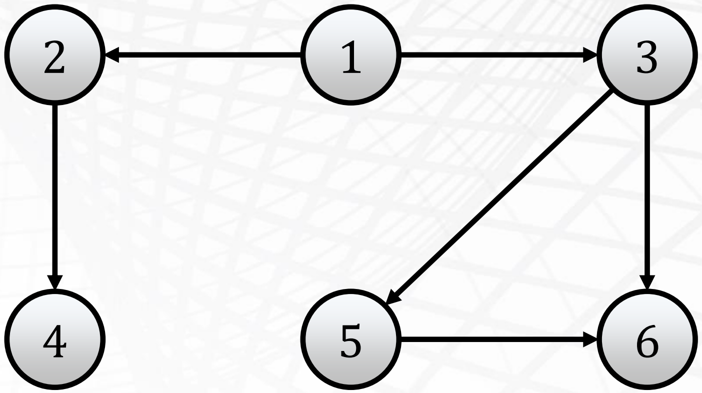
** n.b., it is possible for leap or rest from each of the different states*



# Graph Representation, Revisited

```
vertex(1).
vertex(2).
vertex(3).
vertex(4).
vertex(5).
vertex(6).
edge(1,2).
edge(2,4).
edge(1,3).
edge(3,5).
edge(5,6).
edge(3,6).
```

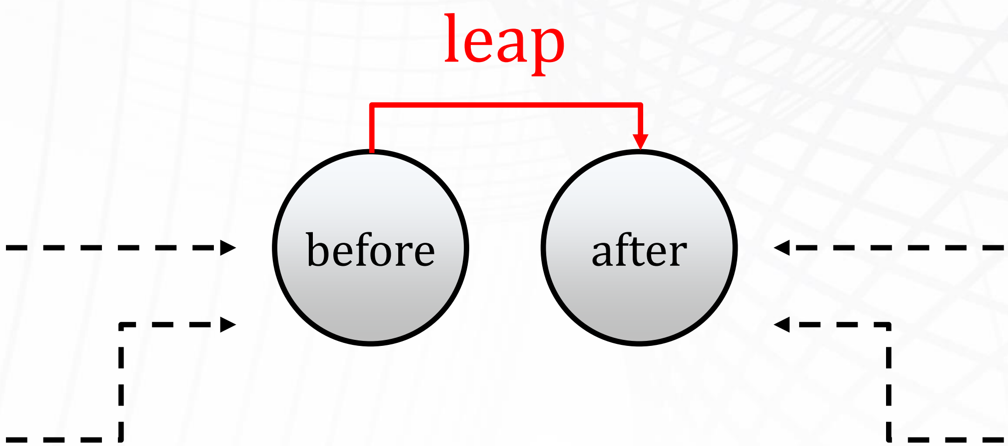



**When Representing** a **Graph** in **Prolog**, **Representing** the **Vertices** often is **Not Required**

**How** do you write a **Generic Tuple** for **Describing** the **Different Transitions** (i.e., **Rest** and **Leap**)?

What is the Time of Day?

Where is the Frog?




What is the Time of Day?

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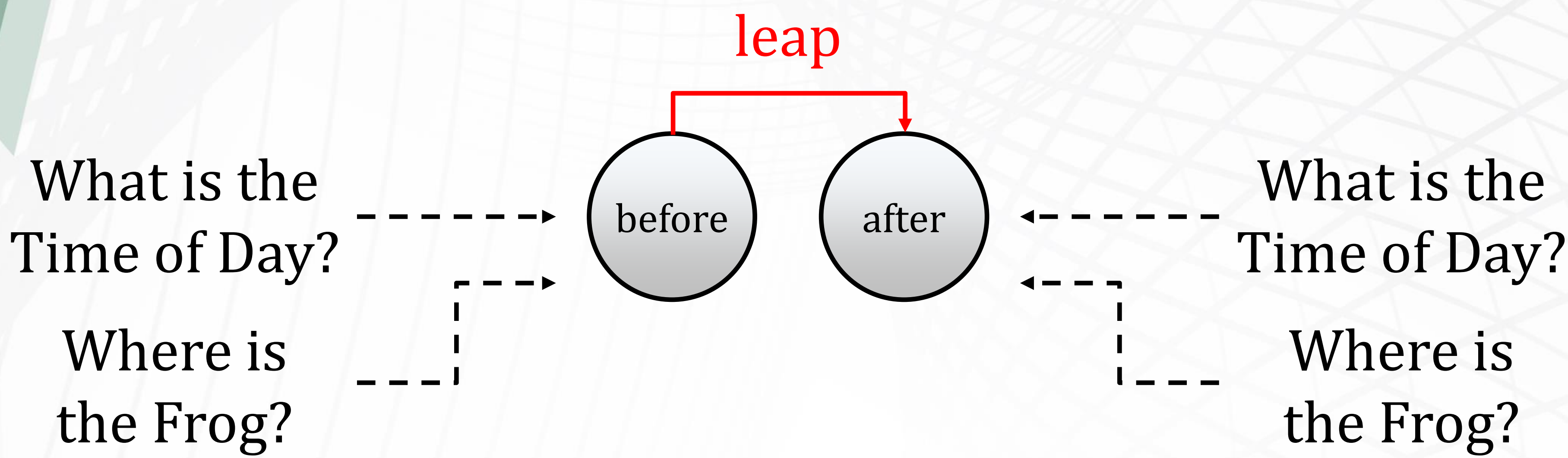
**How** do you write a **Generic Tuple** for **Describing** the **Different Transitions** (i.e., **Rest** and **Leap**)?




```
transition(state(A, B), leap, state(C, D)) :-
    timeMarchesOn(B, D), B \= night, C is A-3.
```

# "Fallen Frog"

```
timeMarchesOn(day, night).
timeMarchesOn(night, day).

transition(state(A, B), leap, state(C, D)) :-
  timeMarchesOn(B, D), B \= night, C is A-3.

transition(state(A, B), rest, state(C, D)) :-
  timeMarchesOn(B, D), C is A+2.

escapeTheWell( state(A, _), [] ) :- A =< 0.

escapeTheWell( state(A, B), [X | Y] ) :- A > 0,
transition( state(A, B), X, C ), escapeTheWell( C, Y ).
```

# "Monkey and Banana"

*A monkey stands by the door in a room.*
*A banana is hanging from a rope in the center of the ceiling.*
*The monkey cannot reach the banana from the floor.*
*There is a window in the room.*
*There is a box sitting under the window.*
*The monkey can move freely around the room.*
*The monkey can move the box.*
*The monkey can climb on top of the box.*
*If the box was under the banana and the monkey was standing on it,*
*then the monkey would be able to reach the banana.*

**Is** it **Possible** for the **Monkey** to get the **Banana**? **How**?

# "Cannibals and Missionaries"

*Three missionaries and three cannibals must cross a river using a boat.*
*Every person is initially on the eastern bank.*
*The boat is initially on the eastern bank as well.*
*The boat can carry at most two people.*
*The missionaries on either bank can never be outnumbered by the cannibals.*
*The boat cannot cross the river without at least one person to guide it.*

**Is** it **Possible** for **Everyone** to **Cross Safely**? **How**?